

⚠ LEG 2 • AUTOMOTIVE AI

THE AI ROAD TRIP

8 more self-driving-car AI systems. Built in 8 days. Zero jargon required.

Swipe to take Exit 01 →



01

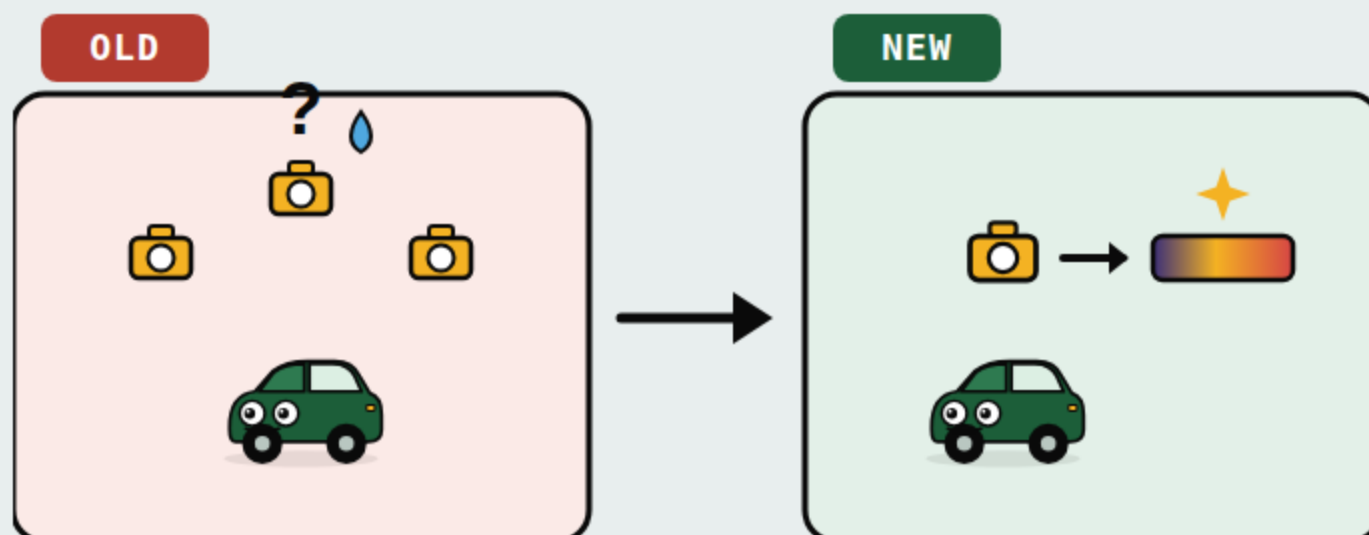
EXIT 01 • DEPTH FROM ONE PHOTO

Borrows an artist's eye to judge distance

This AI repurposes an image-generating model — built to imagine pictures — and retrains it to guess how far away every pixel is, from a single ordinary photo.



Like asking a portrait painter, who already understands shading and perspective, to also tell you exactly how many meters away each object sits.



3D distance from one 2D photo

Marigold

02

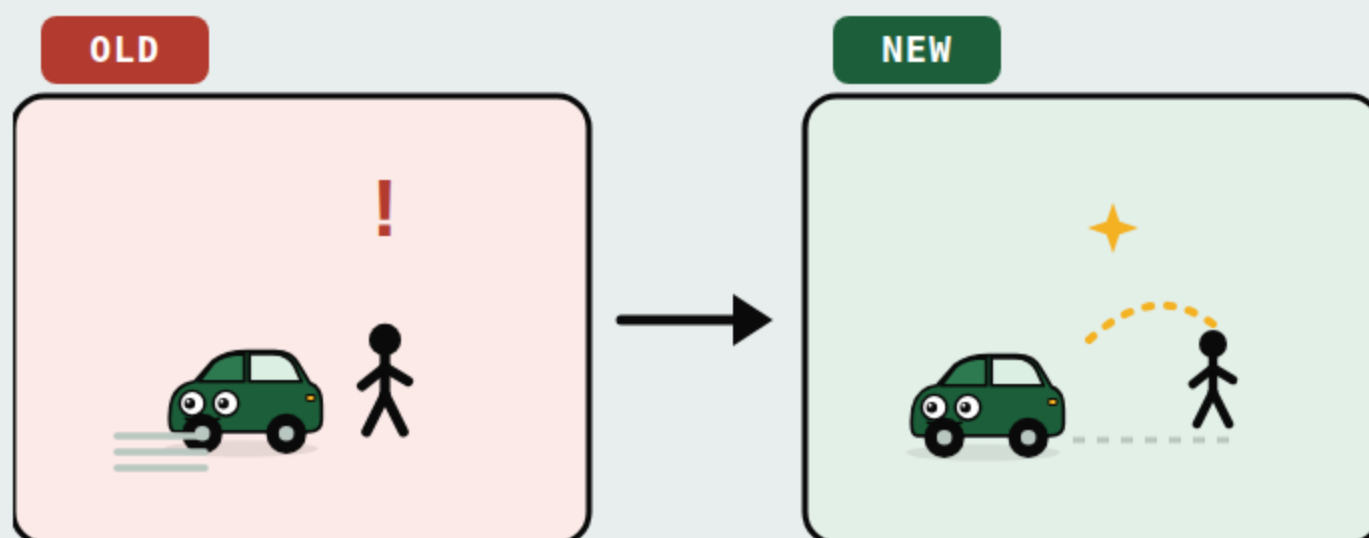
EXIT 02 • PEDESTRIAN INTENT

Reads a pedestrian's body language

This AI watches how someone is walking AND what's around them — a crosswalk, a gap in traffic — to guess if they're about to step into the road, before they do.



Like a driver who notices a pedestrian glance at traffic and slow their stride, and already knows what's coming.



Fewer surprise pedestrian stops

TrajFusionNet+

03

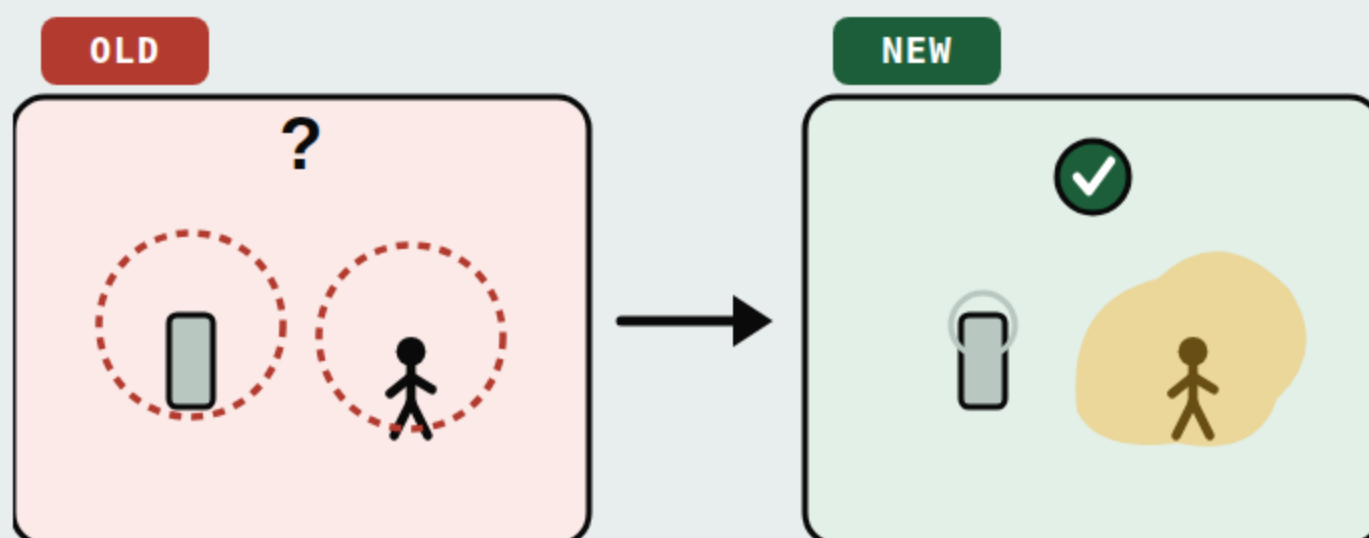
EXIT 03 • RISK-AWARE PLANNING

Trades a fixed danger bubble for a felt one

Old systems draw the same safety circle around every obstacle. This AI learns where danger actually concentrates, and can gently nudge the car's planned path away from it.



Like a driver who senses tension near a distracted pedestrian, not just measures the distance to them.



Shape-shifting safety zones, not fixed circles

READ

04

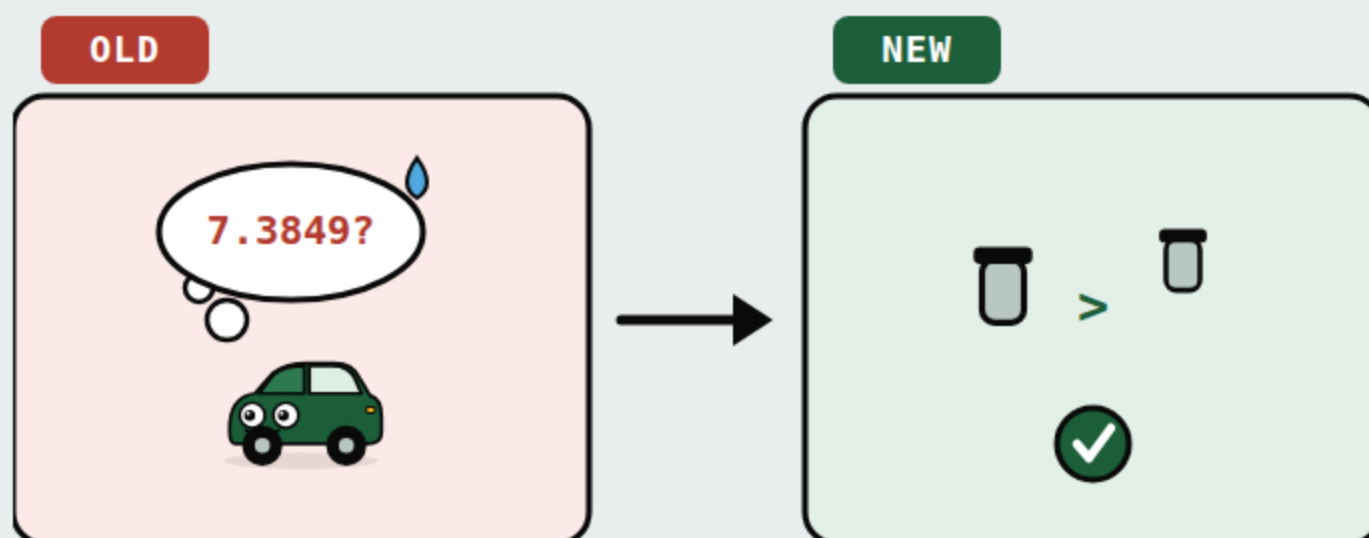
EXIT 04 • RISK RANKING

Doesn't score danger — just ranks it

Instead of learning an exact danger number for every spot on the road, this AI only has to learn one simpler thing: which of two spots is riskier. That turns out to be much easier to get right.



Like a chef who can always tell which of two dishes needs more salt, without ever naming the amount.



+7pt driving score in closed-course testing

DRiF

05

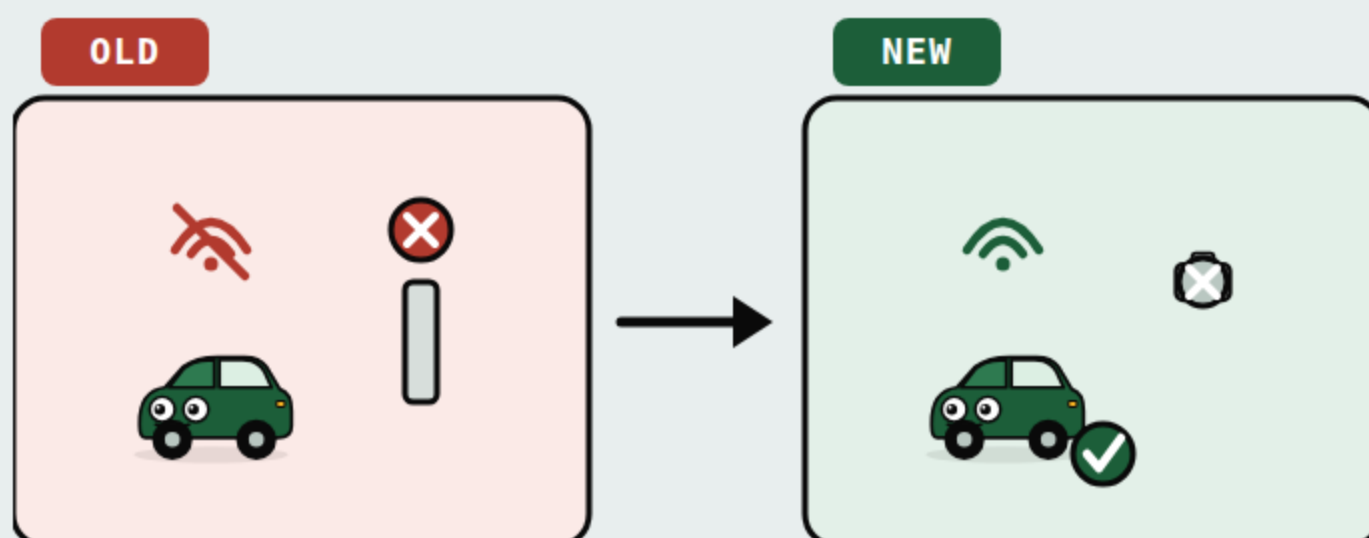
EXIT 05 • V2X CONNECTIVITY

Fills in the blank when a sensor drops out

This AI points a car's wireless connection using cameras, radar, LiDAR and GPS together — and if one sensor goes dark, it invents a believable stand-in instead of losing the signal.



Like finishing a sentence for a friend whose voice just cut out on a call.



Steadier connections, fewer dropped signals

BeamTransFuser

06

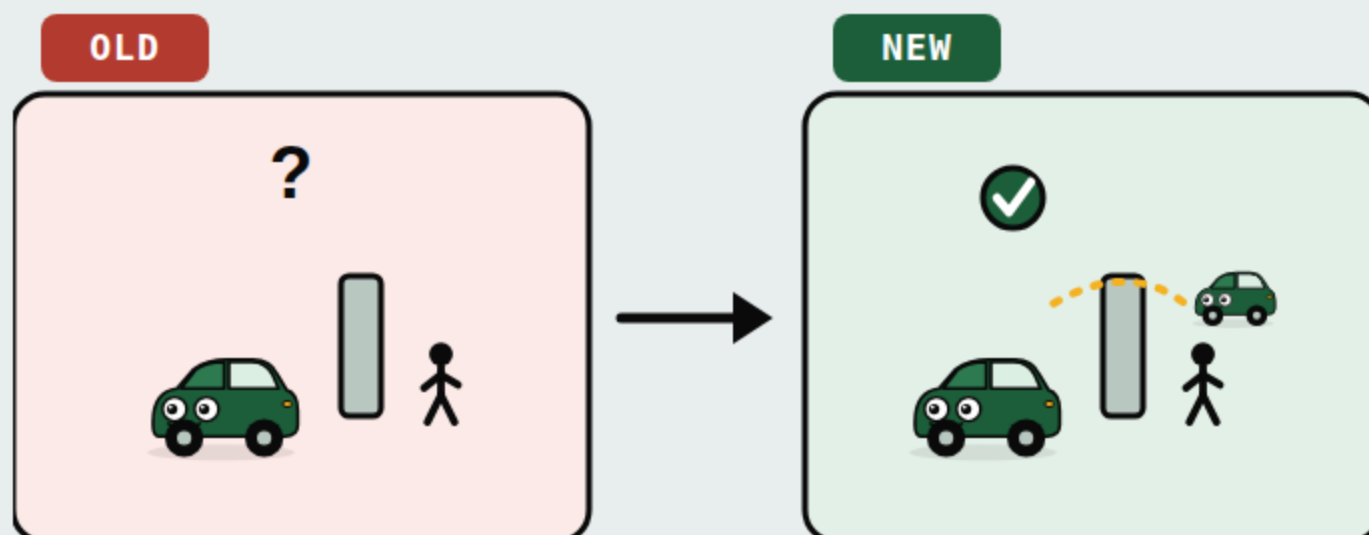
EXIT 06 • CAR-TO-CAR LEARNING

Learns to see better by copying a neighbor

A nearby car shares a compressed glimpse of what it sees over a limited connection, and this AI uses that as a stand-in teacher — no human labels needed — to quietly get better at its own job.



Like learning a new city from a friend's directions, then getting sharper at navigating it yourself.



Cars that teach each other to see

LDE

07

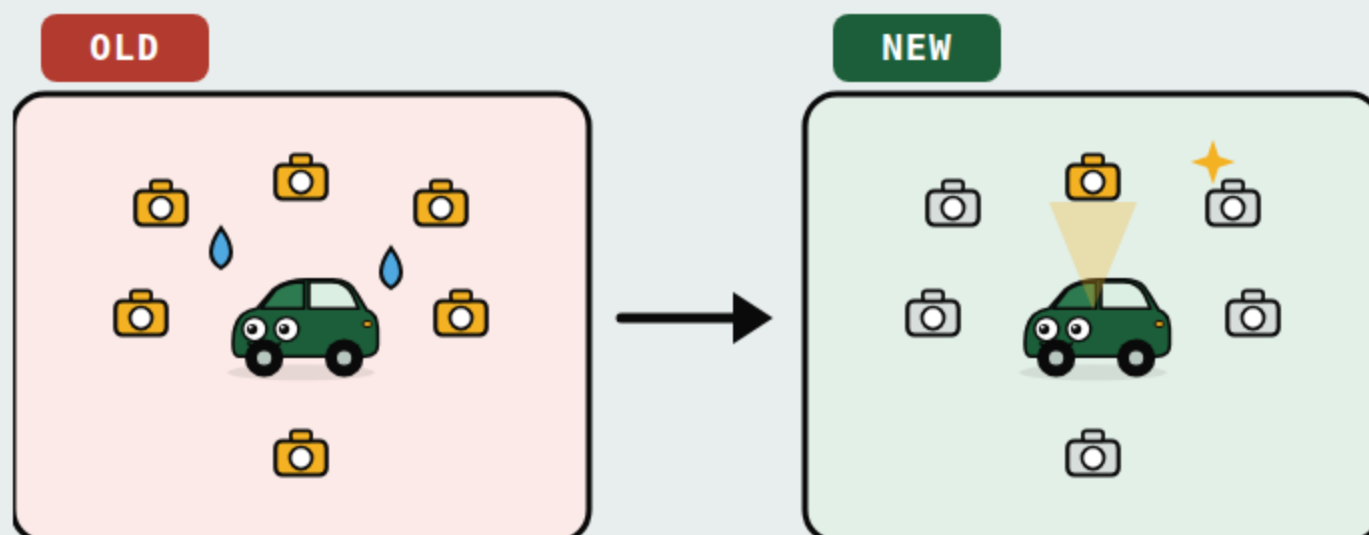
EXIT 07 • EFFICIENT PERCEPTION

Looks everywhere, really checks what matters

Instead of scanning all 6 camera views with equal effort, this AI learns to route its attention only to the regions most likely to hide something — cutting the compute cost of full 360° awareness.



Like a lifeguard who scans the whole pool, but really focuses where someone's struggling.



360° awareness, lighter compute budget

Sparse-BEVNet

08

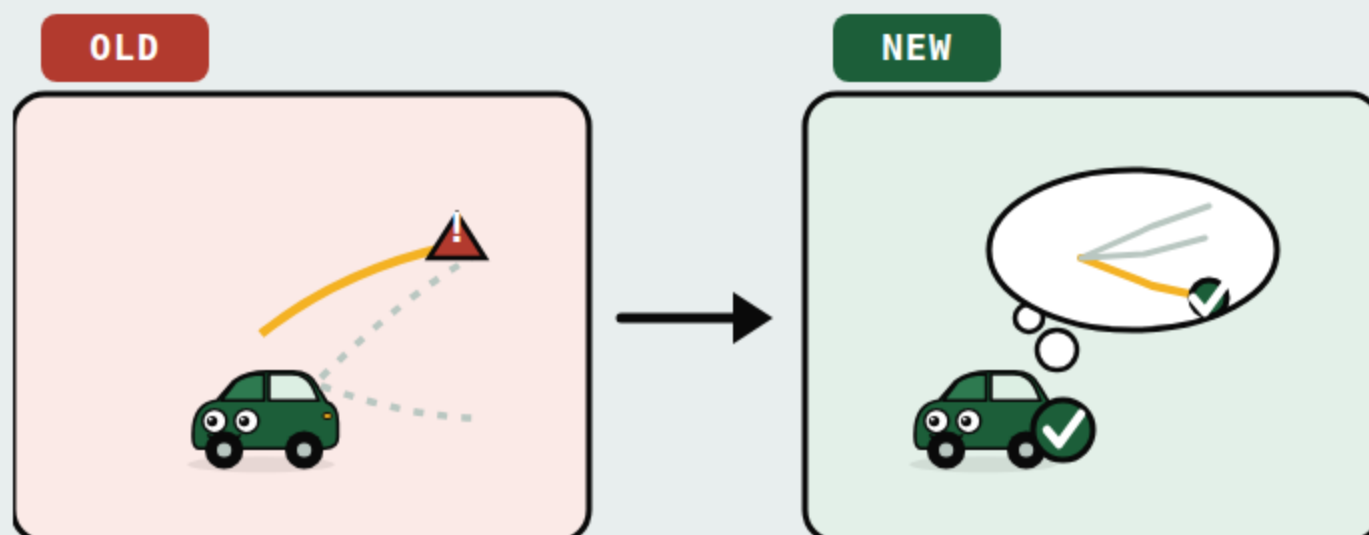
EXIT 08 • IMAGINED FUTURES

Imagines a few futures before picking one

This AI doesn't just plan a path — it dreams up several different next moves AND what the world would look like after each, then picks the move whose imagined future looks safest.



Like a chess player who doesn't just consider a move, but plays out the whole board in their head first.



Plans that consider their own consequences

MM-Future

Want the code?

Full write-ups, tested PyTorch implementations, and simulation GIFs for all 15 systems so far — free and open on GitHub.

`github.com/Lalithprabu/
automotive-ai-daily-reviews`



See the full repo

